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BCA(Sem-VI)–Computer Graphics (602) Core-XIV

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the section as directed.*

Section–A

Multiple choices questions

(Compulsory)

1. Choose the correct answer from the following multiple choice questions : $2\times10=20$

(i) The process through which a vector image is converted into a bitmap image is called :

(a) Pixelization

(b) Quantization

(c) Rasterization

(d) Sampling

(ii) Which among the below set of colours are generally known as the primary colours of light?

(a) White, Yellow, Blue

(b) Red, Green, Black

(c) Red, Green, Blue

(d) Black, white, Red

(iii) Which is not an input device?

(a) Touchscreen

(b) Keyboard

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(c) Mouse

(d) Plotter

(iv) Multimedia contains?

(a) Audio

(b) Video

(c) Both (i) and (ii)

(d) None of these

(v) Graphies can be ____

(a) Simulation

(b) Drawing

(c) Movies, photographs

(d) All of the above

(vi) CAD Stands for :

(a) Computer art design

(b) Computer-aided design

(c) Car art design

(d) None of these

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(vii) The process of positioning an object along a straight line path from one coordinate point to another is called :

(a) Translation

(b) Reflection

(c) Shearing

(d) Transformation

(viii) Which of the following transformation is used for altering the Object's size ?

(a) Translation

(b) Scaling

(c) Rotation

(d) None of these

(ix) The Cohen-Sutherland algorithm divides the two- dimensional space in how many regions?

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(a) 4

(b) 8

(c) 9

(d) 23

(x) DDA stands for ______

(a) Direct Differential Analyzer

(b) Data Differential Analyzer

(c) Direct Difference Analyzer

(d) Digital Differential Analyzer

Section–B

Short answer type questions

Answer any **four** questions of the following :

5×4=20

2. What is computer Graphics ?

3. Explain CRT with diagram.

4. What is clipping? Explain with example.

5. Explain Sutherland and Cohen Line Clipping Algorithm.

6. What are the various area filling techniques?

7. What is a B-spline curve ?

8. Define translation with example.

9. Explain advantages and disadvantages of DVST.

Section–C

Long answer type questions

Answer any **two** questions of the following :

15×2=30

10. Explain the application of computer Graphics.

11. Apply the sutherland and cohen line clipping algorithm to clip the line segment with coordinates (30,60) and (60, 25).

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12. Explain DDA Line drawing algorithm consider the line from (0,0) to (4,6) use the simple DDA algorithm to rasterize this live.

13. Differentiate between vector sean and raster scan display.

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